MULTIMODAL AI TUTORING SYSTEM

25-26J-130

Project Proposal Report

Anjana Chathurangi Kuruppuarachchi

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science

Sri Lanka Institute of Information Technology
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Declaration page of the candidates & supervisor

We declare that this is our own work and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Kuruppuarachchi A.C	IT22334060	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor: Date: 27/08/25.

Signature of the Co Supervisor: Date:

Abstract

The increasing demand for personalized and emotionally responsive education has led to the exploration of artificial intelligence (AI) in intelligent tutoring systems (ITS). This research proposes a multimodal AI-powered tutoring framework centered around a reinforcement learning (RL) engine that dynamically adapts instructional strategies based on real-time learner feedback. Unlike traditional rule-based tutors, the system integrates computer vision (CV), large language models (LLMs), and agentic planning to interpret facial expressions, gaze, posture, and interaction patterns, enabling nuanced understanding of student engagement and emotional state.

The methodology involves capturing multimodal input during live tutoring sessions, extracting semantic and affective features, and fusing them into a structured learner state. The RL engine receives this state and selects optimal instructional actions using a custom reward function that balances correctness, latency, and emotional stability. Instructional content is generated via LLMs and aligned with curriculum goals through agentic planning. All decisions and learner responses are logged for evaluation and policy refinement.

Anticipated results include improved learner engagement, retention, and responsiveness compared to heuristic baselines. The system is expected to demonstrate scalability, ethical data handling, and real-time adaptability across diverse educational contexts. Recommendations will focus on deploying the framework in classroom and remote learning environments, with emphasis on privacy compliance and pedagogical alignment.

Keywords: Reinforcement Learning, Intelligent Tutoring System, Multimodal Feedback, Student Engagement, Adaptive Instruction

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List Of Abbreviations

Abbreviations	Description
AI	Artificial Intelligence
APIs	Application Programming Interface
CV	Computer Vision
DQN	Deep Q-Network
ITS	Intelligent Tutoring Systems
LLMs	Large Language Models
LSTM	Long short-term memory
MDP	Markov Decision Process
MFCC	Mel-Frequency Cepstral Coefficients
POMDPs	Partially Observable Markov Decision Process
PPO	Proximal policy optimization
RL	Reinforcement Learning
SARSA	State-Action-Reward-State-Action
VLM	Vision-Language Model

1. Introduction

The rapid development of Artificial intelligence (AI) has created new opportunities for individualized learning by allowing systems to adjust to the requirements, preferences, and actions of specific students. Reinforcement learning (RL), which defines learning as a sequence of decision-making process, is one of the most promising AI paradigms for educational applications. RL agents, unlike supervised learning agents, learn optimal actions by interacting with the environment and rewards or penalties as feedback. This makes RL particularly suitable for intelligent tutoring systems (ITS), in which the goal is to guide students down complex learning routes while maximizing engagement and retention [1], [2].

In traditional ITS, instructional decisions are often based on static rules or heuristics, which fail to capture the dynamic and nonlinear nature of human learning. RL offers a more flexible and data-driven approach, allowing tutors to adapt in real time based on student performance, emotional state, and engagement levels [3]. However, despite its theoretical advantages, RL in education faces several practical challenges, most notably, the lack of real-world deployment, limited action spaces, and insufficient integration of multimodal feedback [4], [5].

This proposal introduces a strategic RL engine as a core component of a Multimodal AI Tutoring System, designed to overcome these limitations. The engine will be trained on both simulated and live student data, using a custom reward function that incorporates performance metrics, attention cues, and emotional signals. By integrating with Agentic AI planning, LLM-driven content generation, and CV-based attention monitoring, the RL engine will enable proactive, personalized, and pedagogically sound instruction [6], [7].

1.1 Background & Literature Survey

The application of RL in education has gained momentum in recent years, with researchers exploring various algorithms, environments, and pedagogical strategies. A comprehensive review by Wang et al. [2] analyzed over 50 studies, categorizing them by RL method (e.g., Q-learning, SARSA, DQN, policy gradients), domain (e.g., math, language learning), and student environment (simulated vs. live). The review found that RL agents consistently outperform heuristic tutors in simulations, achieving higher retention and mastery rates. However, only a small fraction of studies involved real students, highlighting a significant gap between theory and practice. A more recent systematic review by Riedmann et al. [8] examined 89 RL-in-education studies and emphasized the need for broader deployment and evaluation with actual learners. The authors identified key challenges in affective modeling, policy stability, and ethical implementation. One notable system is RLTutor, developed by Singh and Kumar [3], which used Q-learning with epsilon-greedy exploration to optimize hint timing and content sequencing. The tutor was trained on virtual students modeled using knowledge tracing and demonstrated superior performance compared to rule-based systems. However, the system was tested only in simulation, and its effectiveness in real classrooms remains unknown. Another study by Zhang et al. [4] introduced a Smart E-Learning Framework that applied Q-learning to recommend personalized learning paths in MOOC environments. The framework modeled learning as an MDP and used simulated user feedback to evaluate performance. While the system improved engagement and path efficiency, it lacked emotional modeling and real-world validation. Kim et al. [5] proposed a multi-agent RL framework with experience sharing, enabling tutor agents to exchange state-transition data and accelerate policy convergence. This approach was promising for resource-limited environments, where single agents had limited knowledge. Scalability and issues regarding privacy were not fully addressed, however, and the system was only tested in simulation environments.

A more recent study by Fahad et al. [9] also categorized RL techniques in education, i.e., deep RL, POMDPs, and hybrid models, and emphasized the need to integrate multimodal feedback to enable real-time adaptivity. Similarly, Chen and Lee [6] wrote about multimodal AI for learning with an emphasis on the role of visual reasoning and sketch-based interaction towards enhancing

learner engagement. Common tools across such studies are Python-based libraries for RL (e.g., OpenAI Gym, TensorFlow, PyTorch) and datasets such as ASSISTments, EdNet, and synthetic learner simulations. Cumulative reward, retention, mastery rate, duration of engagement, and policy convergence time are common metrics of evaluation.

1.2 Research Gap

Despite growing publications, several significant gaps remain in the application of RL to intelligent tutoring systems:

• Real-World Deployment: Most RL-based tutors remain confined to lab experiments with minimal validation in real classrooms. This restricts their generality and real-world applicability [2], [3], [8].

• Narrow Action Spaces: Recent models focus mainly on limited pedagogical actions, e.g., hinting or content ordering, but do not consider broader pedagogical strategies like pacing, motivation, or affective support [4], [9].

• Lack of Multimodal Feedback Incorporation: Few systems incorporate real-time affective or attentional feedback from learners, crucial for adaptive learning. The absence of CV or sensor-based information limits the tutor to react to non-verbal cues [5], [6], [10].

• Privacy and Ethical Concerns: Collection and processing of sensitive information of students, especially emotional or behavioral cues, raises ethical and privacy issues that must be addressed through secure and transparent systems [2], [9].

• Computational Complexity: Scalable infrastructure and efficient algorithms are required to learn RL agents on heterogenous large student data. Convergence rate and resource usage are the challenges faced by most current systems [5], [8].

These limitations highlight the need for a robust multimodal RL engine that can learn in real classrooms, react to diverse learner profiles, and play nicely with other AI building blocks.

1.3 Research Problem

The central research question addressed in this proposal is:

How can a reinforcement learning engine be designed to dynamically adapt instructional strategies in real-time, using multimodal feedback to reduce long-term learning gains across different learning environments?

This question encompasses various sub-problems:

• Designing a reward function weighing short-term engagement and long-term retention and mastery.

• Learning RL agents on both simulated and real student data, with generalization over learning styles.

• Embedding visual and affective cues within the state space such that the tutor can respond to non-verbal cues.

• Ensuring scalability, efficiency, and privacy conformance in actual implementations.

Solving this problem will enable the development of next-generation intelligent tutoring systems that are not just reactive but also proactive, strategic, and emotionally intelligent.

2. Objectives

2.1 Main Objective

To design and develop a multimodal AI-powered intelligent tutoring system that enhances personalized learning by integrating reinforcement learning, large language models, computer vision, and real-time feedback mechanisms—aimed at improving student engagement, emotional responsiveness, and long-term academic performance.

This objective represents the collective goal of the project: building a responsive and adaptive tutoring system that understands learners holistically and supports them effectively.

2.2 Specific objectives

- To design a reinforcement learning engine that dynamically adapts instructional strategies based on student performance, attention, and emotional cues.
- To use large language models (LLMs) to generate tailored learning content, explanations, and feedback tailored to meet individual student needs.
- To integrate computer vision (CV) techniques for student attention tracking and facial expressions, offering real-time emotional sensitivity and engagement measurement.
- To develop a multimodal feedback system that consolidates visual, auditory, and behavioral information to inform instructional decision-making.
- To ensure ethical handling of data and privacy compliance throughout the entire system, especially when handling sensitive emotional and behavioral markers.
- To evaluate the system's performance in simulated and authentic learning environments, quantifying performance measures such as retention, engagement, and adaptability.
- In order to measure the scalability and effectiveness of the whole tutoring system to run in various educational settings and learner profiles.

3. Methodology

3.1 Overall System Diagram

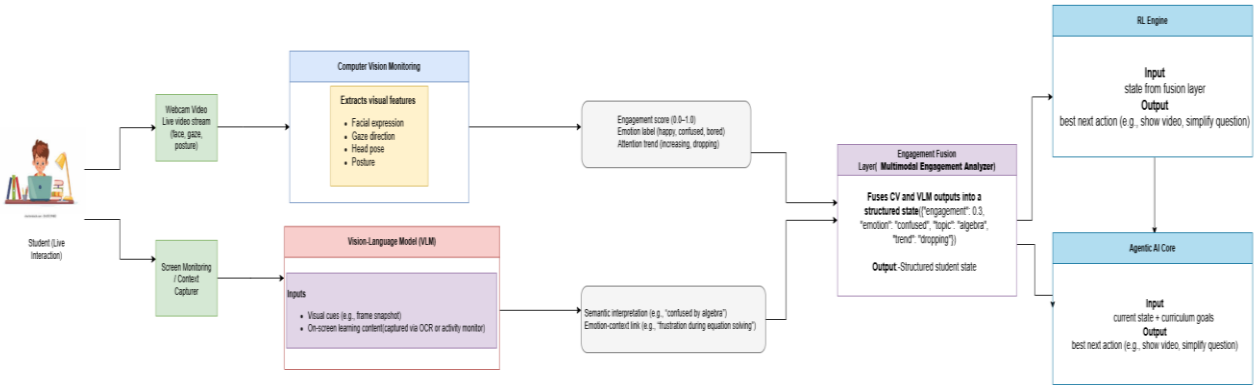

Figure 1 System Architecture

The planned intelligent tutoring system is an in-real-time, multimodal AI system that is designed to offer adaptive, affectively intelligent instruction. The major objective is to encourage personalized learning through the integration of reinforcement learning (RL), computer vision (CV), large language models (LLMs), and agentic planning.

As shown in the system diagram, the learner is presented with the system through a webcam and tutor interface. The on-screen activity and webcam feed are processed by CV and Vision-Language Model (VLM) modules to extract gaze, emotion, and engagement signals. They are fused into a structured learner state that includes attention trends, emotional labels, and engagement scores.

This learner state is then transmitted to the Agentic AI Core, which constructs teaching objectives and produces prompts. The LLM Content Generator generates personalized quizzes, cues, and explanations. Meanwhile, the RL Engine monitors reward signals (e.g., correctness, latency, engagement) and selects the optimal teaching action. All learner responses and choices are monitored for analysis and policy update.

The architecture is a closed-loop system, continually adapting to the learner's cognitive and affective state and mapping to curricular objectives. Ethics and privacy are maintained by a dedicated Privacy and Policy Guard, which manages consent management, data storage, and access control.

3.2 Individual System Diagram

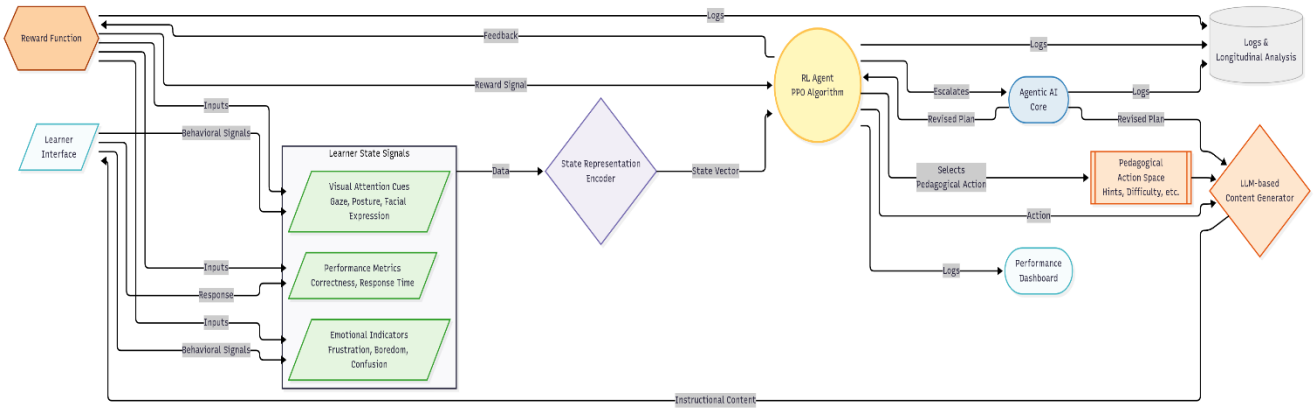

Figure 2 RL Engine System Architecture

The RL engine is used to be the adaptive decision-making core of the intelligent tutoring system. Based on the real-time input gathered from the other elements—learner performance, emotional state, and engagement signals—the RL engine decides on the best instructional action to take. Such actions may include offering a hint, adjusting the difficulty level, paraphrasing an explanation, or giving motivational feedback.

The multimodal input to the RL engine from the system is in the form of:

• Performance metrics (e.g., correctness, response time)

• Visual attention cues (e.g., gaze, posture, facial expression)

• Frustration, boredom, confusion signals

These inputs are processed and coded as structured state representation. The RL agent, having been trained on a policy optimization algorithm such as PPO [11], evaluates the state and selects

an action from a provided pedagogical action space. The selected action is then sent to the LLM-based content generator, which supplies the appropriate instruction material to the learner.

If the student shows disengagement or emotional disturbance, the RL engine prioritizes actions that are designed to re-engage the student—e.g., making the content less complex, offering encouragement, or modality shift (e.g., visual explanation instead of written form). If the student is performing well and remains engaged, the RL engine can raise the level of complexity or introduce challenge-based assignments in order to sustain momentum.

The reward function the RL engine uses is a blend of:

• Task achievement (e.g., correct answers, quicker time to mastery)

• Indicators of engagement (e.g., sustained attention, reduced dropout risk)

• Emotional steadiness (e.g., reduced frustration signals, positive affect)

The RL engine updates and learns its policy continuously through feedback gained from the learner's responses and emotional signals. The feedback loop enables the system to gain experience and sharpen its instructional strategy over time. Where the RL engine recognizes chronic disengagement or incompetent learning (e.g., persistent mistake-making, affective burnout), it is able to send the issue up the Agentic AI Core, which can revise the long-term learning plan or suggest a change in instructional approach. It is thus attuned not only to short-term interaction but also to longer-term learning trajectories. All the decisions made by the RL engine are tracked and displayed in a performance dashboard so that instructors and researchers can monitor system behavior, learner progress, and policy learning. The dashboard can also be used to enable longitudinal analysis for improving the reward function and increasing overall adaptivity in the system.

3.3 Project Execution Strategy

The project will be carried out in five phases:

1. System Design: Define architecture, data flow, and module interfaces.
2. Component Development: Build and train CV, VLM, RL, and agentic planning modules.
3. System Integration: Connect all components into a unified platform.
4. Pilot Deployment: Test the system with simulated and real learners.
5. Evaluation and Refinement: Analyze performance and improve adaptivity.

3.4 Tasks and sub tasks

1. Multimodal Input Capture and Pre-Processing

- Integrate video conferencing and screen-sharing tools
- Set up audio and video capture using libraries like Librosa and OpenCV
- Build preprocessing modules for noise reduction and frame alignment
- Synchronize audio, video, and text inputs for unified analysis

2. Feature Extraction and Representation

- Extract audio features such as pitch, pace, and energy
- Apply MFCC and spectral techniques for audio feature selection
- Use computer vision to extract facial expressions, gaze, and engagement cues
- Process learner text input using tokenization and embeddings

3. Reinforcement Learning Engine Development

- Design the RL environment and define reward signals based on learner behavior
- Implement the RL agent using suitable algorithms (e.g., PPO, DQN, LSTM)

- Train the agent using simulated learner data and public datasets
- Evaluate and refine the agent's performance across learning scenarios

4. Adaptive Feedback and Agentic Planning

- Integrate LLMs for generating personalized hints and explanations
- Build a planning module to sequence learning tasks based on learner state
- Implement feedback delivery strategies (text, voice, visual cues)
- Monitor feedback effectiveness and adjust strategies accordingly

5. Real-Time Monitoring and Analytics

- Connect model outputs to a real-time dashboard using APIs
- Develop modules for generating learner progress reports
- Visualize engagement and decision metrics using data visualization tools

6. System Integration and Testing

- Combine all modules into a unified system architecture
- Test the system across platforms (web and mobile)
- Optimize for real-time performance and responsiveness
- Ensure ethical data handling and privacy compliance

3.5 Materials and Resources

Hardware: Laptop with webcam for students

Software: Python – Core language for model development, RL training, and system orchestration, MongoDB – Databases for storing learner profiles, interaction logs, and curriculum metadata, Jupyter Notebooks – Prototyping and visualization of RL training, reward functions, and policy behavior, draw.io –system diagram design for documentation

Project Management: Task scheduling tools such as Gantt charts will be used to manage timelines and ensure that each phase is completed on time.

3.6 Data Requirements and Collection

Data Needed

- Webcam video (facial expressions, gaze, posture)
- Screen activity logs (contextual cues)
- Learner interaction data (responses, timestamps)
- Emotional state labels and engagement scores

Collection Methods

- Real-time capture during tutoring sessions
- Synthetic data generation for RL pretraining

By the end of the project, the system is expected to:

- Deliver personalized instructional actions based on multimodal engagement signals
- Improve learner attention, emotional responsiveness, and retention
- Demonstrate scalability and ethical data handling in real-world educational settings
- Serve as a blueprint for adaptive ITS deployment in classrooms, online platforms, and special education contexts

The RL engine will be benchmarked against heuristic baselines and evaluated using engagement metrics, learning outcomes, and educator feedback. Its ability to adaptively select instructional strategies will be validated through pilot studies and longitudinal analysis.

4. Project Requirements

4.1 Functional Requirements

These define the core capabilities the system must perform to fulfill its educational objectives.

- Real-time Engagement Monitoring: Capture and analyze student facial expressions, gaze, posture, and screen activity.
- Multimodal Fusion: Combine visual and contextual data into a structured learner state (engagement score, emotion label, attention trend).
- RL-Based Decision Making: Use reinforcement learning to select optimal instructional actions based on learner state.
- Curriculum Alignment: Agentic AI Core ensures selected actions align with pedagogical goals.
- Feedback Loop: Continuously update learner state and adapt instructional strategy based on new observations.
- Data Logging: Store interaction data for evaluation, model refinement, and ethical auditing.

4.2 Non-Functional Requirements

These define system qualities that support performance, usability, and scalability.

- Performance: Real-time inference latency < 500ms for engagement detection and RL decision-making.
- Privacy & Ethics: GDPR-compliant data handling, anonymization, and consent-based logging.
- Modularity: Each component (CV, VLM, RL, Agentic Core) must be independently testable and replaceable.

- Scalability: Support for multiple concurrent learners in cloud-based deployment.
- Explainability: RL decisions must be interpretable via reward trace and learner state visualization.
- Maintainability: Clear documentation, version control, and automated testing pipelines.

4.3 User Requirements

These describe what users expect from the system.

- Learners should receive personalized, timely feedback during sessions
- Tutors should access learner analytics and engagement summaries
- Admins should manage users, monitor system health, and update content
- All users should experience seamless interaction across devices

5. Gantt Chart

	Task	Start Date	Due Date
1	Research Project	2025/03/24	2026/06/30
2	Brainstorming	2025/03/25	2025/03/26
3	Supervisor selection and group registration	2025/05/26	2025/05/27
4	TAF Submission	2025/06/30	2025/07/16
5	Proposal Presentation	2025/07/08	2025/07/12
6	Proposal Submisson	2025/07/19	2025/07/20
7	50% Completion	2025/12/15	2026/12/19
8	PP1	2025/12/15	2026/12/19
9	80% Completion	2026/03/01	2026/06/30
10	PP2	2025/03/01	2026/06/30
11	Final Presentation	2025/05/01	2026/06/30
12	Final report submisson	2025/06/01	2026/06/30

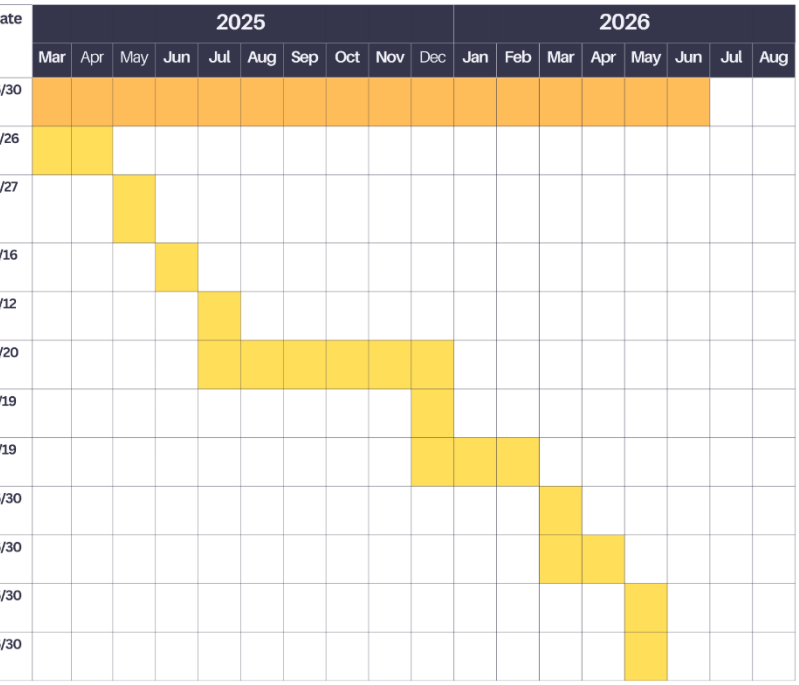

Figure 3 Gantt Chart

5.1 Work Breakdown Structure

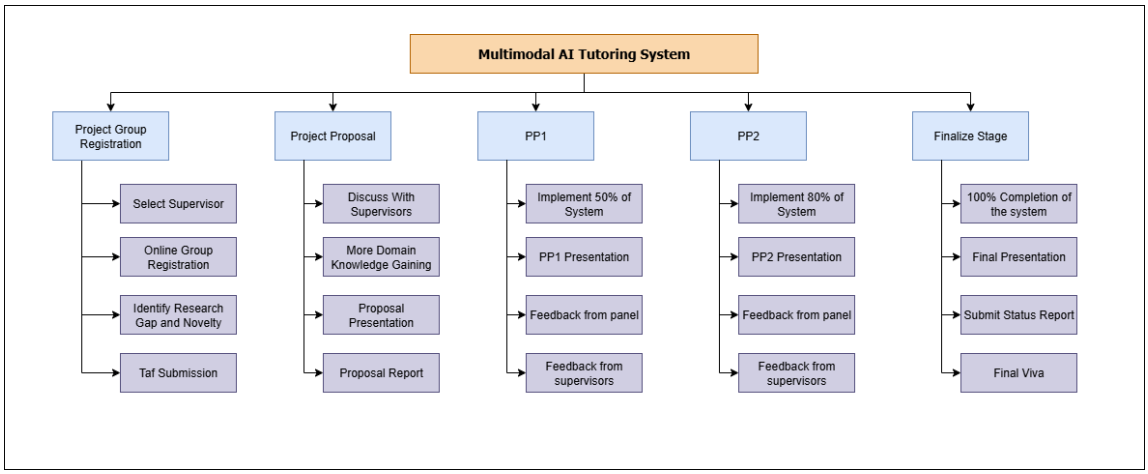

Figure 4 Work Breakdown Structure

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